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 What is Generative AI?

Day 1 — Notebook 1 of 3 | Estimated Time: 30 minutes

What You'll Learn

- The difference between AI, Machine Learning, and Generative AI
- How Generative AI differs from traditional AI
- Real-world applications of Generative AI
- The current Generative AI landscape and key players

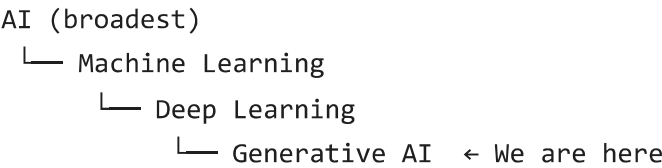
1. AI → ML → Deep Learning → Generative AI

Let's break down the hierarchy:

Term	What It Means	
Artificial Intelligence (AI)	Any system that can perform tasks that normally require human intelligence	Chess er
Machine Learning (ML)	AI systems that <i>learn from data</i> rather than being explicitly programmed	Recomm
Deep Learning	ML using neural networks with many layers	Image re
Generative AI	Deep Learning models that can <i>create new content</i> (text, images, code, audio)	ChatGP1

Key Insight

Traditional AI **classifies** or **predicts** (e.g., "Is this email spam?"). Generative AI **creates** (e.g., "Write me an email about...").



2. How Does Generative AI Work? (The 10,000-Foot View)

Generative AI models are trained on massive datasets of text, images, or other content. Through this training, they learn **patterns** — the statistical relationships between words, pixels, or tokens.

When you give a model a prompt, it doesn't "think" or "understand" in the human sense. Instead, it **predicts the most likely next token** based on the patterns it learned during training.

The Core Process:

- 1. **Training:** The model reads billions of documents and learns patterns
- 2. **Prompting:** You give the model an input (a prompt)
- 3. **Generation:** The model predicts tokens one at a time to create a response

 **Think of it like autocomplete on steroids.** Your phone's keyboard predicts the next word — LLMs do the same thing, but with much more sophistication and context.

3. What Can Generative AI Do?

Here are some real-world applications across different domains:

Domain	Application	Example
Text	Writing, summarization, translation	Drafting emails, summarizing reports
Code	Code generation, debugging, explanation	GitHub Copilot, code review assistants
Conversations	Chatbots, customer support	AI assistants, customer service bots
Data	Data analysis, extraction, transformation	Parsing invoices, extracting entities
Search	Semantic search, Q&A over documents	RAG-powered knowledge bases
Images	Image generation, editing	Marketing creatives, product mockups
Audio	Speech synthesis, music generation	Voice cloning, podcast generation

4. The Generative AI Landscape

Key Model Families

Provider	Model	Strengths
Google	Gemini (1.5, 2.0, 2.5)	Multimodal, long context windows, function calling
OpenAI	GPT-4, GPT-4o	Strong reasoning, code generation
Anthropic	Claude 3, 3.5	Safety-focused, long context
Meta	LLaMA 3	Open-source, community-driven
Mistral	Mistral, Mixtral	Efficient open-source models

In This Course

We'll focus on **Google Gemini** models, accessed through the `google-generativeai` Python SDK. Gemini models are:

- **Multimodal** — they can process text, images, audio, and video
- **Available in different sizes** — Flash (fast & cheap), Pro (powerful), Flash-Lite (smallest & fastest)
- **Accessible via API** — no GPU needed, just an API key

5. What is a Large Language Model (LLM)?

An LLM is a type of Generative AI model specifically designed to work with **text**. The "Large" refers to the number of parameters (weights) in the model — modern LLMs have billions to trillions of parameters.

Key Characteristics of LLMs:

Property	Description
Parameters	Billions of learned weights (e.g., GPT-4 has ~1.8 trillion)
Training Data	Trained on large portions of the internet, books, code
Context Window	How much text the model can "see" at once (e.g., Gemini 2.5 supports up to 1M tokens)
Token-Based	Models work with tokens, not words (more on this in Notebook 2)
Probabilistic	Outputs are <i>sampled</i> — the same prompt can give different outputs

What LLMs Are Good At:

- ✓ Text generation, summarization, translation
- ✓ Question answering and reasoning
- ✓ Code generation and debugging
- ✓ Following complex instructions

What LLMs Struggle With:

- ✗ Factual accuracy (they can "hallucinate")
- ✗ Real-time information (training data has a cutoff)
- ✗ Math and precise calculations
- ✗ Accessing external systems without tools

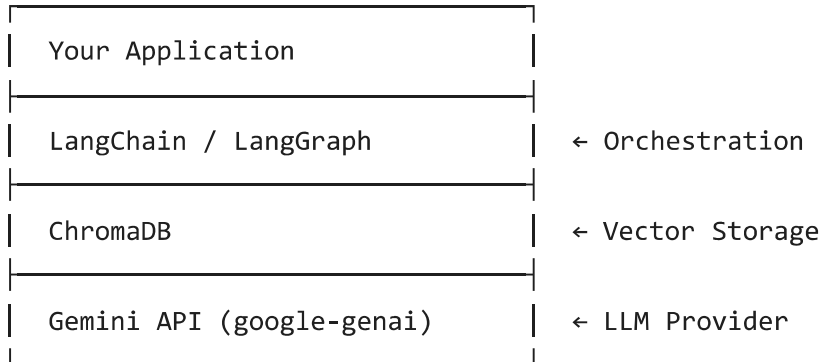
6. How We'll Use LLMs in This Course

Throughout this program, you'll learn to:

1. **Prompt Engineer** — Write effective prompts to get the best outputs from LLMs

2. **Build RAG Systems** — Augment LLMs with external knowledge via retrieval
3. **Evaluate Outputs** — Systematically measure LLM quality
4. **Build Agents** — Create LLM-powered agents that can use tools and reason

The Stack We'll Use:



✓ Exercise 1: Identify the AI Type

For each of the following applications, classify whether it is **Traditional AI**, **Machine Learning**, or **Generative AI**:

1. A system that detects fraudulent credit card transactions
2. A chatbot that writes marketing copy for products
3. A rule-based system that routes customer support tickets
4. A model that generates realistic images from text descriptions
5. A recommendation engine that suggests movies on Netflix

Write your answers in the cell below:

```
# Exercise 1: Classify each application
# Replace the "???" with: "Traditional AI", "Machine Learning", or "Generative

answers = {
    "1. Fraud detection": "Machine Learning",
    "2. Marketing copy chatbot": "Generative AI",
    "3. Rule-based ticket routing": "Traditional AI",
    "4. Text-to-image generation": "Generative AI",
    "5. Movie recommendations": "Machine Learning",
}

for item, answer in answers.items():
    print(f"{item}: {answer}")
```

1. Fraud detection: Machine Learning
2. Marketing copy chatbot: Generative AI
3. Rule-based ticket routing: Traditional AI
4. Text-to-image generation: Generative AI
5. Movie recommendations: Machine Learning

✓ Exercise 2: Strengths and Weaknesses

For each task below, would you trust an LLM to do it well? Write **"Yes"**, **"No"**, or **"With Safeguards"** and a brief reason why.

1. Summarize a 50-page research paper
2. Calculate your company's quarterly tax obligation
3. Draft a customer support email response
4. Provide today's stock price for Apple
5. Explain a Python error traceback

```
# Exercise 2: Can you trust an LLM for this?
# Fill in your answers below
```

```
trust_assessment = {
    "1. Summarize research paper": {"verdict": "Yes", "reason": "It is pretrain"},
    "2. Calculate quarterly tax": {"verdict": "No", "reason": "Taxes are subjec"},
    "3. Draft support email": {"verdict": "Yes", "reason": "They are good in pr"},
    "4. Today's stock price": {"verdict": "No", "reason": "They don't work well"},
    "5. Explain Python error": {"verdict": "Yes", "reason": "They are good in e"}
}

for task, assessment in trust_assessment.items():
    print(f"{task}")
    print(f" Verdict: {assessment['verdict']}")
    print(f" Reason: {assessment['reason']}")
    print()
```

1. Summarize research paper
Verdict: Yes
Reason: It is pretrained with 1B+ parameters and hence can predict next words
2. Calculate quarterly tax
Verdict: No
Reason: Taxes are subject to dynamic changes with respect to governments
3. Draft support email
Verdict: Yes
Reason: They are good in predicting new words
4. Today's stock price
Verdict: No

Reason: They don't work well with real time data

5. Explain Python error

Verdict: Yes

Reason: They are good in elaborating python errors , unless a new type of err



Further Reading & Videos

Resource	Type	Link
Intro to Large Language Models	Video (1hr)	Andrej Karpathy — YouTube
What is Generative AI?	Article	Google Cloud
State of AI Report	Report	stateof.ai
AI Index Report (Stanford HAI)	Report	aiindex.stanford.edu

Next up: [02_How_LLMs_Work.ipynb](#) — Deep dive into Transformers, attention, and tokens